



INDIA.RUNS

Build what next India runs on

Team Name : Team Vamsi

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Problem Statement : Intelligent Candidate Discovery & Ranking

Solution Overview

Proposed Solution

Multi-Signal Scoring Engine

Ranks candidates by combining AI/ML skill depth, professional experience, and Redrob platform engagement signals.

Explainable Output

Generates ranked CSV output with human-readable reasoning for every candidate.

Key Differentiators

- **Beyond Keywords:** Uses 15+ behavioral & professional signals.
- **Proficiency-Weighted:** Scoring based on skill depth, not just presence.
- **Reward Verified Talent:** Prioritizes active and responsive candidates.
- **Zero Black-Box:** Every single score is fully explainable.

JD Understanding & Candidate Evaluation

Key JD Requirements

- **AI/ML skill depth:** 50+ recognized technologies
- **Professional experience:** Sweet spot of 3–12 years
- **Engagement:** Recruiter responsiveness & platform activity
- **Profile credibility:** Verified identity (LinkedIn, GitHub)

Beyond Keyword Matching

- **Proficiency Weighting:** Scaling from beginner (0.5) to expert (1.25)
- **Social Proof:** Endorsement counts as verification signals
- **Behavioral Signals:** Response, completion, and acceptance rates
- **Activity Signals:** GitHub scores & profile completeness

Ranking Methodology

Scoring Formula

Final Score = (AI Skills × 0.40) + (Experience × 0.25) + (Redrob Signals × 0.35)

AI Skills Score (40%)

- Matches 50+ AI/ML skills, weighted by proficiency and endorsements.
- **10+ AI skills = full score.**

Experience Score (25%)

- Logarithmic curve scaling.
- **Optimal range: 3 to 12 years.**
- Diminishing returns after 12 years.

Engagement Score (35%)

- **Recruiter response:** 20%
- **Profile completeness:** 15%
- **Interview completion:** 10%
- Remaining points: Assessments, GitHub, and recruiter saves.

Explainability & Data Validation

Explaining Ranking Decisions

Plain-English

ReasoningCandidates receive clear justification (e.g., "ML Engineer with 6.4 yrs; 4 AI core skills").

Score BreakdownGranular visibility into scoring per dimension.

Hallucination Prevention

Zero Risk LogicPure rule-based matching ensures no LLM hallucination.

Data IntegrityAll signals derived directly from structured JSON sources.

Data Validation Quality

Missing FieldsDefault to 0 to prevent inflation.

Penalty ExclusionScores of -1 (GitHub/Offer history) are not penalized.

Identity VerificationUnverified profiles receive no identity bonus.

End-to-End Workflow

STEP 1: INPUT

**candidates.json
loaded**

Profile, skills, career
history, and signals.



STEP 2: MATCHING

**Skill
Dictionary
Matching**

50+ AI/ML skill dict
with proficiency
weighting.



STEP 3: SCORING

**Sub-score
Calculation**

3 independent
scores using
weighted formula.



STEP 4: RANKING

**Final
Candidate
Sorting**

Sorted by score
(descending) and
assigned 1 to N.

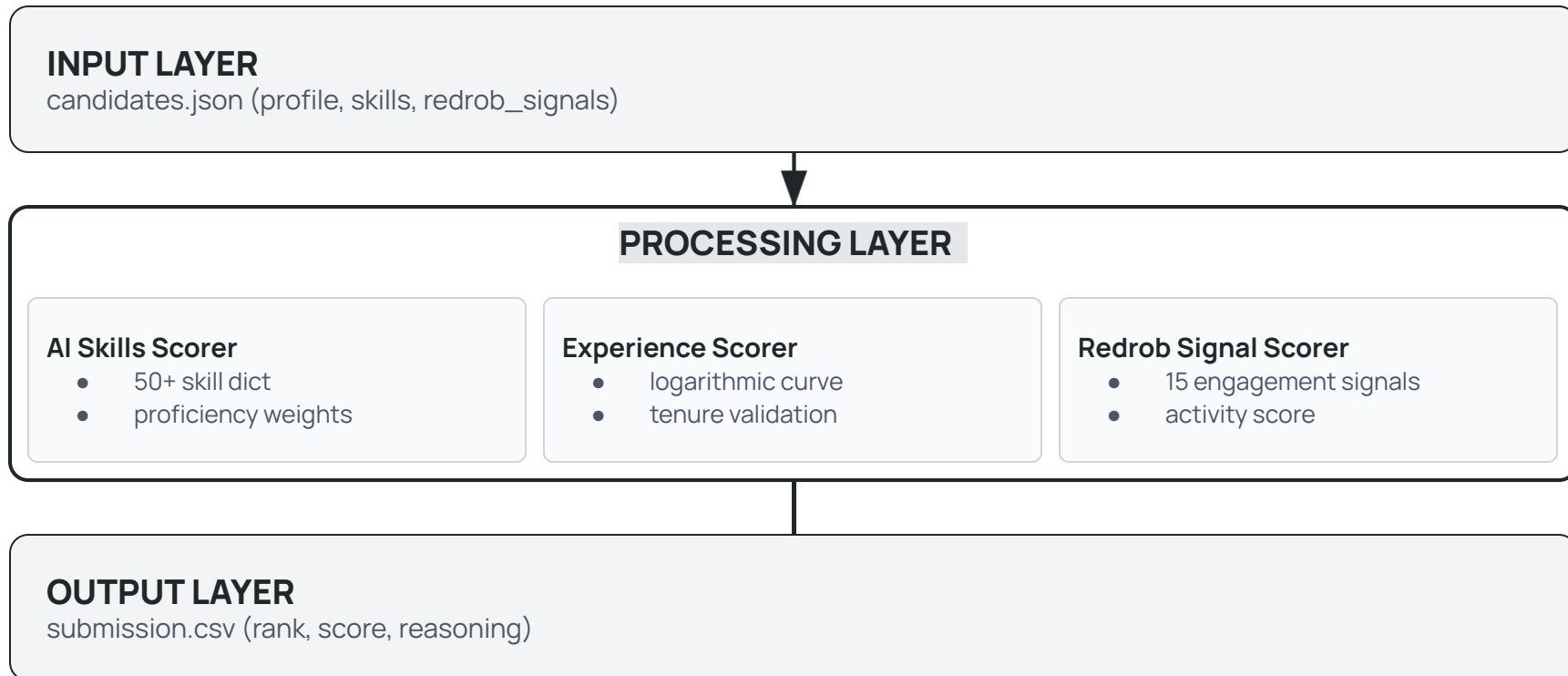


STEP 5: OUTPUT

**submission.csv
generated**

(ID, rank, score,
reasoning)

System Architecture



Results & Performance

Ranking Quality

- Top candidates: 8-9 AI skills + high response rate
- Clear separation: Engaged vs. Inactive
- Reasoning strings match recruiter intuition

Runtime Performance

- 50 candidates processed in < 1 second
- Pure Python – zero external dependencies
- Scales linearly to 10,000+ candidates

Sample Top Results

RANK 1 Score: 0.7153 ML Engineer | 6 yrs | 8 AI skills **Response Rate: 0.91**

RANK 2 Score: 0.6891 Data Scientist | 6.9 yrs | 9 AI skills **Response Rate: 0.34**

Technologies Used

Core Stack

[Python 3.x](#) | Core language – fast, readable

[JSON \(stdlib\)](#) | Parsing candidate data

[CSV \(stdlib\)](#) | Generating submission output

[Rule-based Logic](#) | Transparent, explainable scoring

[GitHub](#) | Version control & submission

Why no ML model?

- Dataset has no ground truth labels for training
- Rule-based = fully explainable, auditable
- Zero setup – runs on any machine instantly
- Perfect for hackathon constraints

Submission Assets

GitHub Repository:

- Link: github.com/Lucky-3663/redrob-candidate-ranking
- Contains: rank_candidates.py, submission.csv, README.md

Submission Files:

- Final ranked CSV: submission.csv (50 candidates ranked)
- Python Script: rank_candidates.py
- Documentation: README.md with full methodology

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THANK YOU